

Q.1) Which of the following measurements represents the reading of the micrometer?

☒ a) 9.25 mm

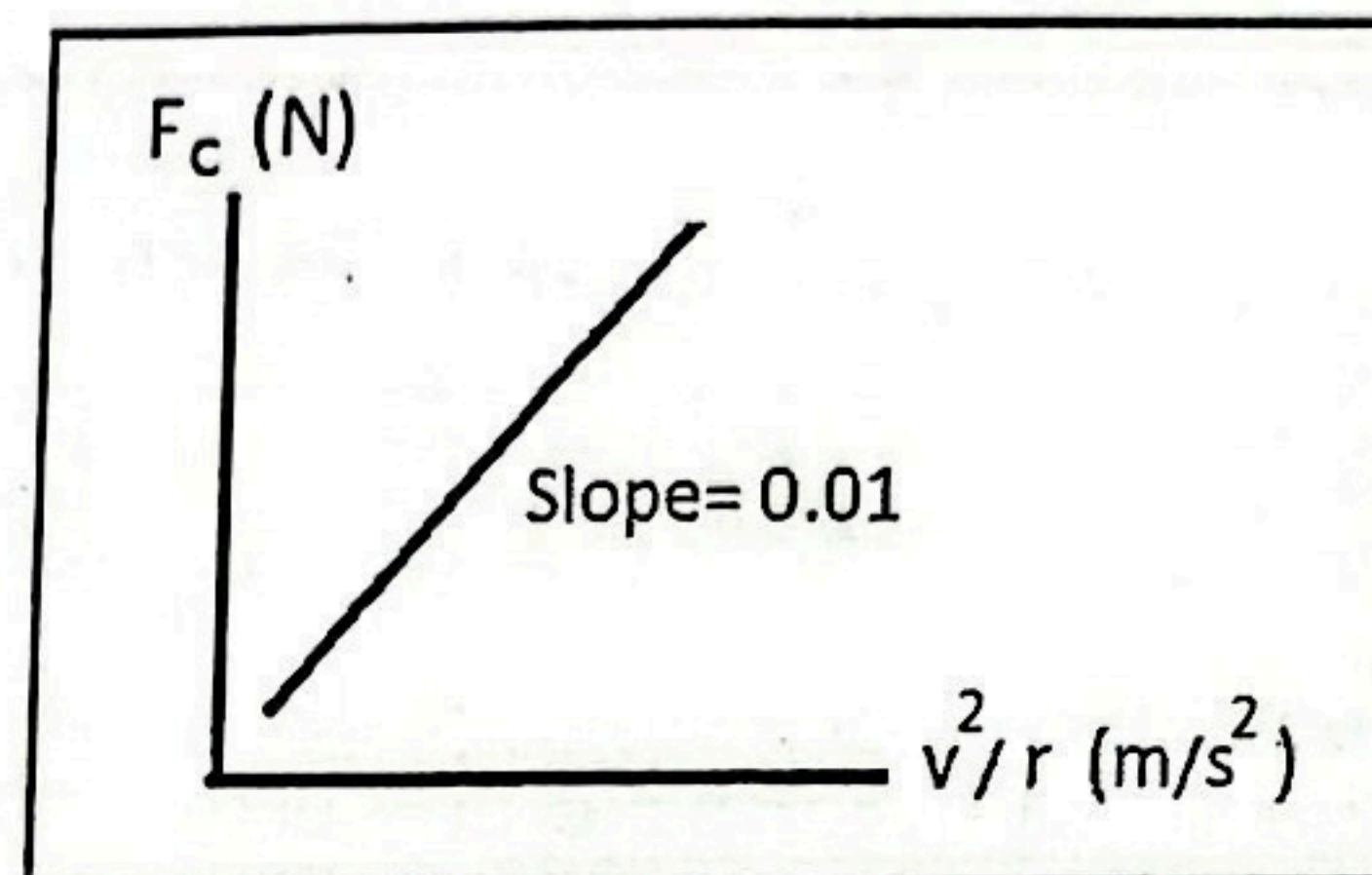
b) 8.0 cm

c) 1.50 cm

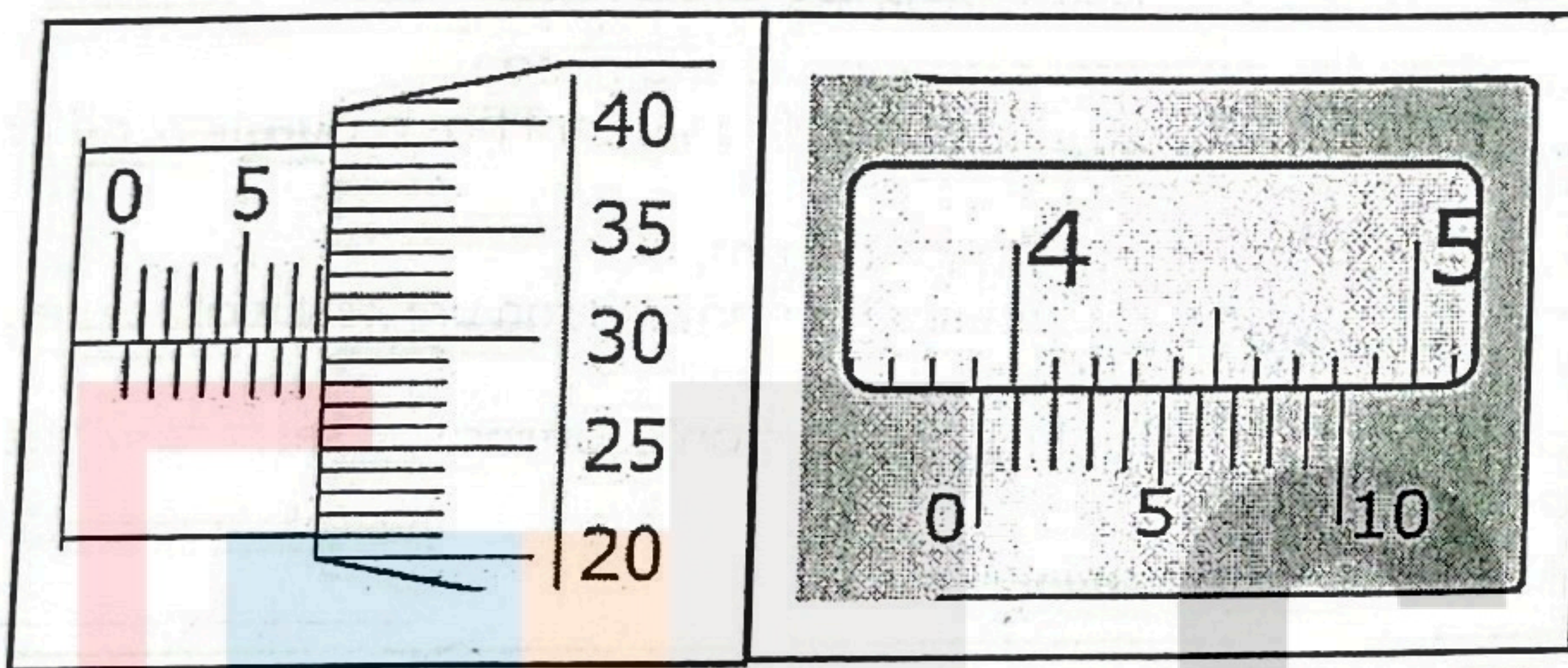
d) 2 mm

Q.2) In uniform circular motion experiment, the radius (r) = 0.4 m. If we got this graph shown, the mass (m) is:

- ☒ a. 10 g
 b. 11 g
 c. 100 g
 d. 0.1 g



Q.3) Using significant figures rules, find the multiplying of these readings:



- a. 325.4 mm² b. 330 mm² c. 113 mm² ☒ d. 325 mm²

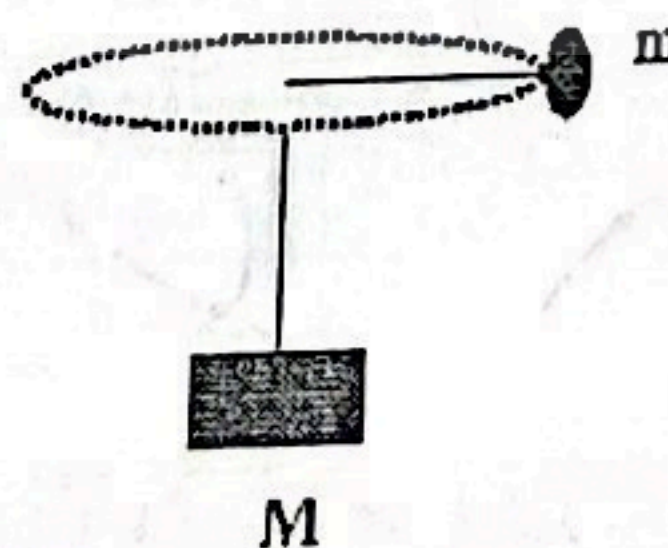
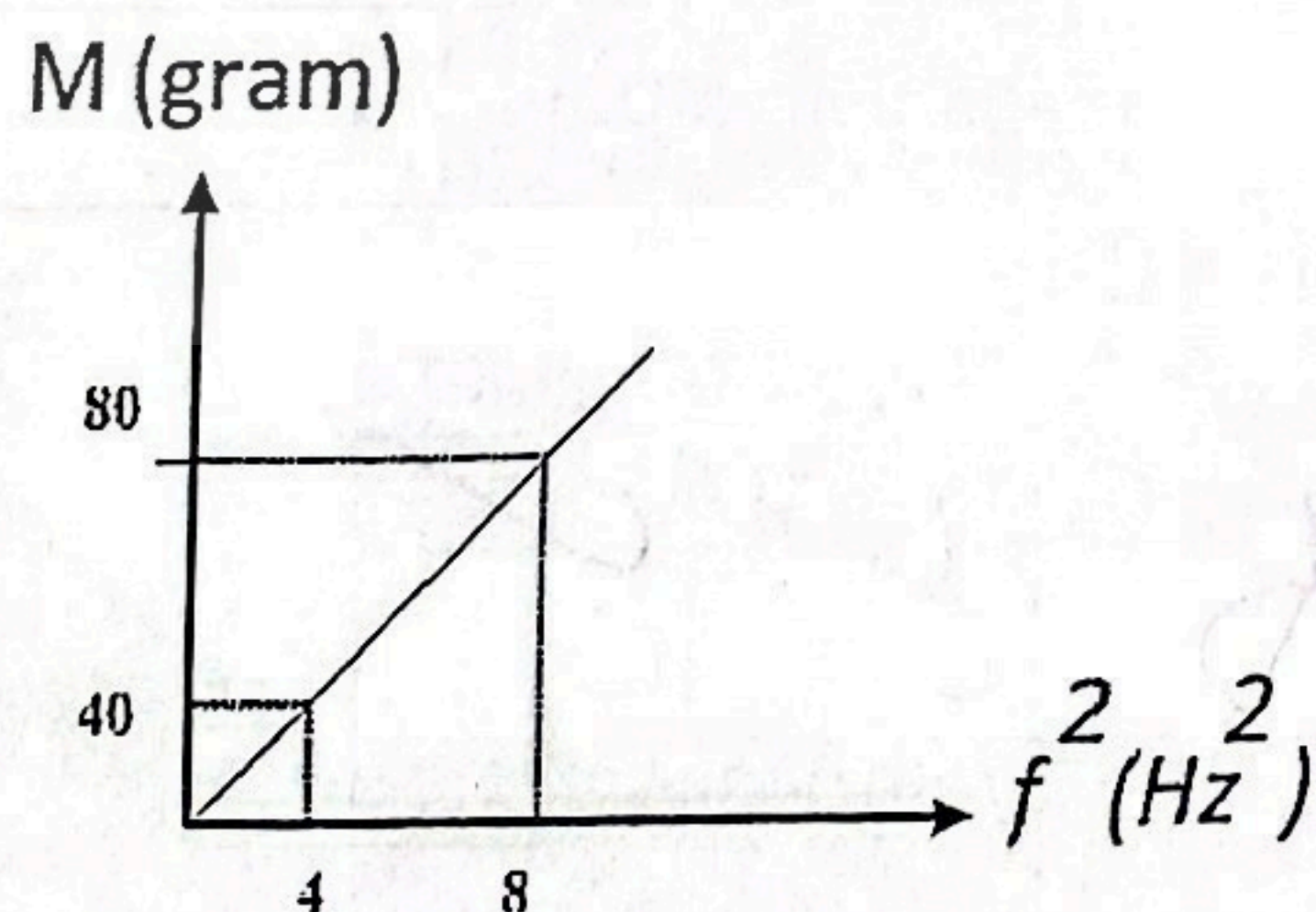
Q.4) In vector experiment, the ring mass does not effect on the equilibrium state.

- a. True ☒ b. False

Q.5) The best equipment to measure the thickness of the paper is the vernier caliper

- ☒ a. True b. False

Q.6) In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius ($R=0.3$ m). A plotted data are shown in the figure bellow. ($g=10$ m/s²)



The mass (m) of the rotational body is:

a) 6.34 gram

b) 12.7 gram

c) 8.45 gram

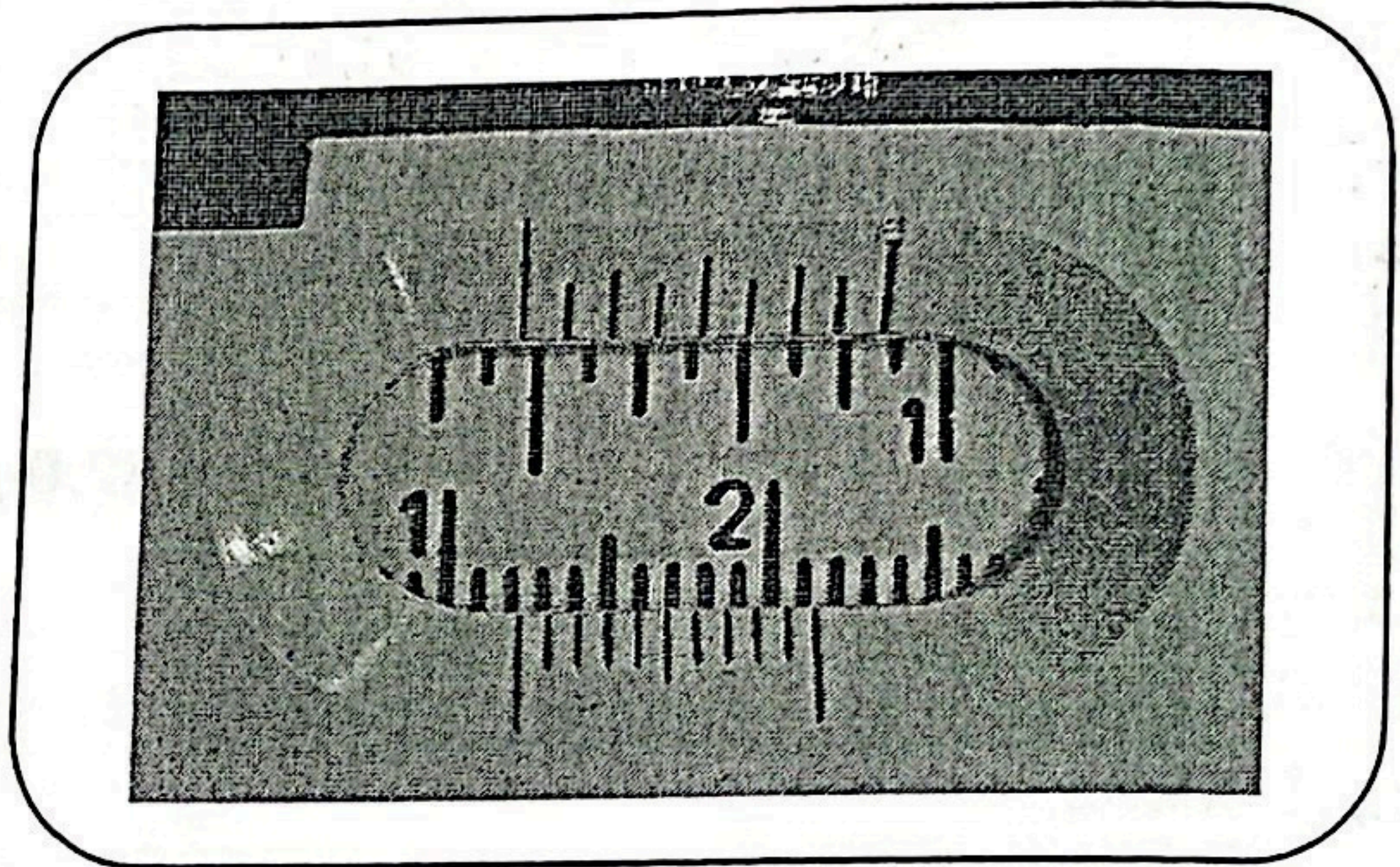
d) 4.22 gram

Q.13) How many significant figures in the number of 100 as in this sentence:
"There are 100 years in a century".

- a. 1 significant digits
- b. 2 significant digits
- c. infinite significant digits
- ☒ d. 3 significant digits

Q.14) what is the readings of the following instruments?

- ☒ a. 12.3 ± 0.05 mm
- b. 1.23 ± 0.005 mm
- c. 123.0 ± 0.05 mm
- d. 1.23 ± 0.0005 mm



Q.15) A cylinder has a radius of $R \pm \Delta R = (2.00 \pm 0.05)$ cm and a length of $L \pm \Delta L = (20.0 \pm 0.1)$ cm. Calculate the volume and the error in the volume of the cylinder $V \pm \Delta V$ in cm^3 ?

- a. 251 ± 7.54
- ☒ b. 251 ± 14
- c. 512 ± 70.5
- d. 351 ± 17.63

Q.7) An object is moving with a constant speed in a circular path of radius r . If its frequency of revolution is $(3n)$, express its centripetal acceleration in terms of n and r .

- a. $4\pi^2 n^2 r$
- b. $9\pi^2 n^2 r$
- c. $3\pi^2 n^2 r$
- ☒ d. $36\pi^2 n^2 r$

Q.8) which of the following measurements represents the reading of the caliber?

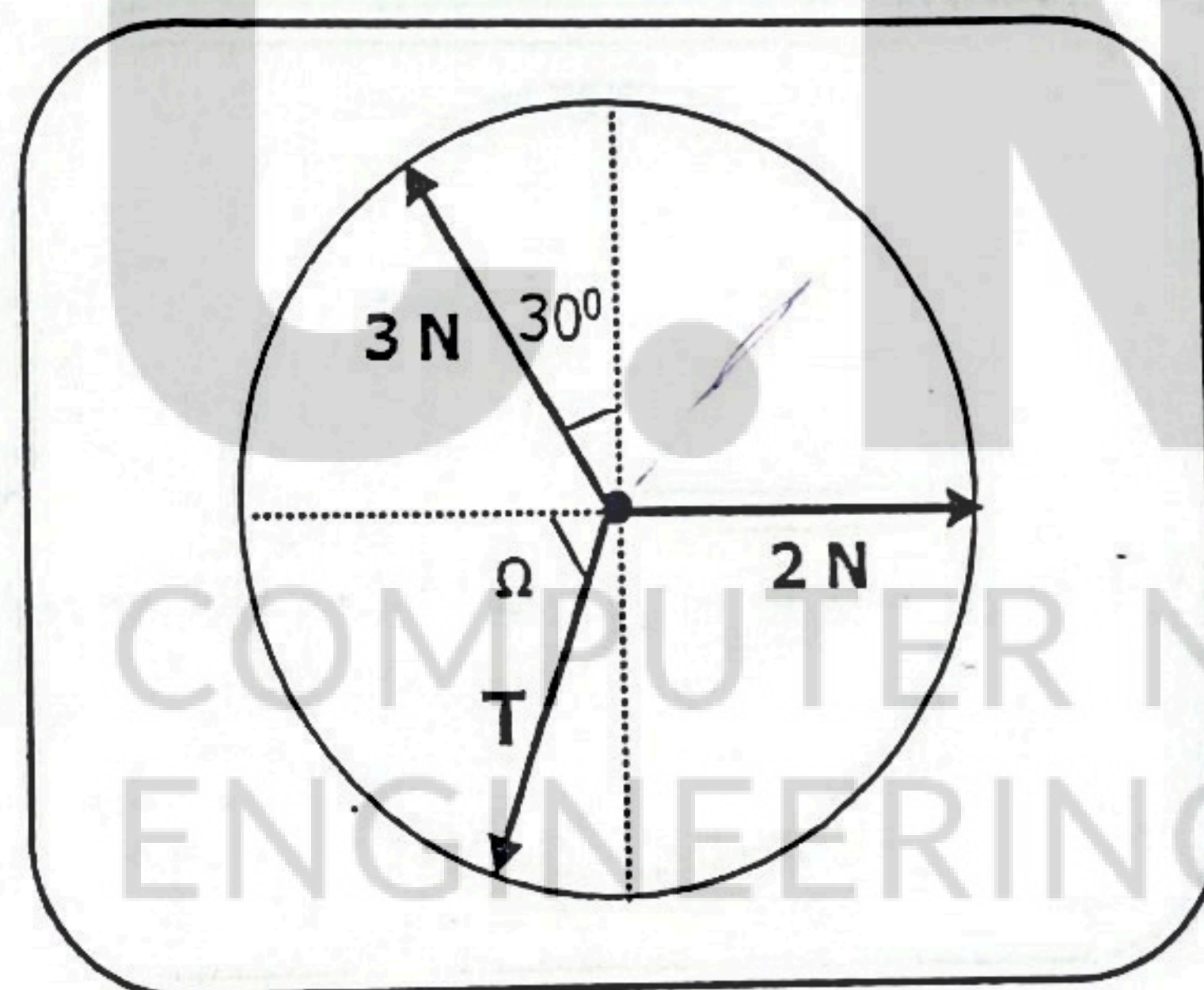
- a. 2 mm
- b. 8.0 cm
- ☒ c. 1.50 cm
- d. 20.0 mm

Q.9) Compute the following: $(0.049623 \times 32.0 / 478.8)$

- a. 0.00331
- b. 0.00330
- ☒ c. 0.00332
- d. 0.00300

Q.10) For the experimental setup shown. Consider a body located at the origin, the body is in equilibrium, if T (in N), and Ω (in degree) are respectively:

- a) 2.64, 79.1
- ☒ b) 2.64, 52.3
- c) 3.27, 52.3
- d) 3.27, 79.1



Q.11) Round **3.78721** to three significant figures

- a. 3.78
- b. 3.77
- ☒ c. 3.79
- d. 3.80

Q.12) Round **24.85** to three significant figures

- a. 24.7
- b. 24.9
- c. 25.0
- ☒ d. 24.8